

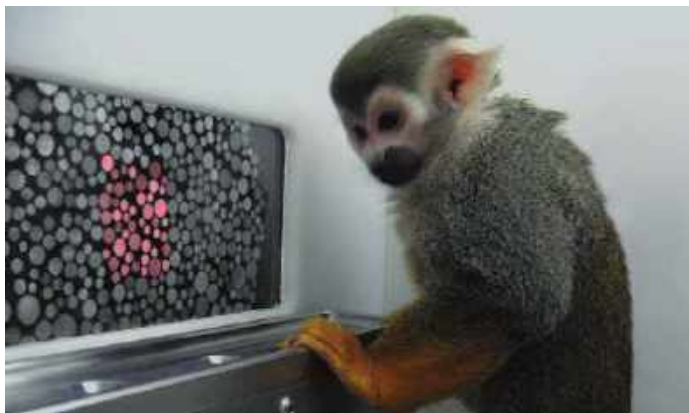
from the binding of retinal to three distinct opsin proteins. Slight differences in the opsin proteins cause each photopsin to absorb light optimally at a different wavelength. Although the visual pigments are designated as red, green, and blue, their absorption spectra in fact overlap. For this reason, the brain's perception of intermediate hues depends on the differential stimulation of two or more classes of cones. For example, if both red and green cones are stimulated, we see either yellow or orange, depending on which class of cones is more strongly stimulated.

Abnormal color vision typically results from mutations in the genes for one or more photopsin proteins. In humans, color blindness nearly always affects perception of red or green and is far more common in one gender than the other: 5–8% of males, fewer than 1% of females. Why? The human genes for both the red and green pigments are X-linked. Thus, males are affected by a single mutation, whereas females are color-blind only if both copies are mutant. (The human gene for the blue pigment is on chromosome 7.)

Experiments on color vision in the squirrel monkey (*Saimiri sciureus*) enabled an early breakthrough in the field of gene therapy. These monkeys have only two opsin genes, one sensitive to blue light and the other sensitive to either red or green light, depending on the allele. Because the red/green opsin gene is X-linked, all males have only the red- or green-sensitive version and are red-green color-blind. When researchers injected a virus containing the gene for the missing version into the retina of adult male monkeys, evidence of full color vision was apparent after 20 weeks (**Figure 50.21**).

The squirrel monkey gene therapy studies demonstrate that the neural circuits required to process visual information can be generated or activated even in adults, making it possible to treat a range of vision disorders. Indeed, gene therapy

▼ **Figure 50.21 Gene therapy for vision.** Once color-blind, this adult male monkey treated with gene therapy demonstrates his ability to distinguish red from green.



MAKE CONNECTIONS Red-green color blindness is X-linked in squirrel monkeys and in humans (see Figure 15.7). Why is the inheritance pattern in humans not apparent in squirrel monkeys?

has been used to treat Leber's congenital amaurosis (LCA), an inherited retinal degenerative disease that causes severe loss of vision. After using gene therapy to restore vision in dogs and mice with LCA, researchers successfully treated the disease in humans by injecting the functional LCA gene in a viral vector (see Figure 20.22).

➔ **Mastering Biology HHMI Video: Genes as Medicine**

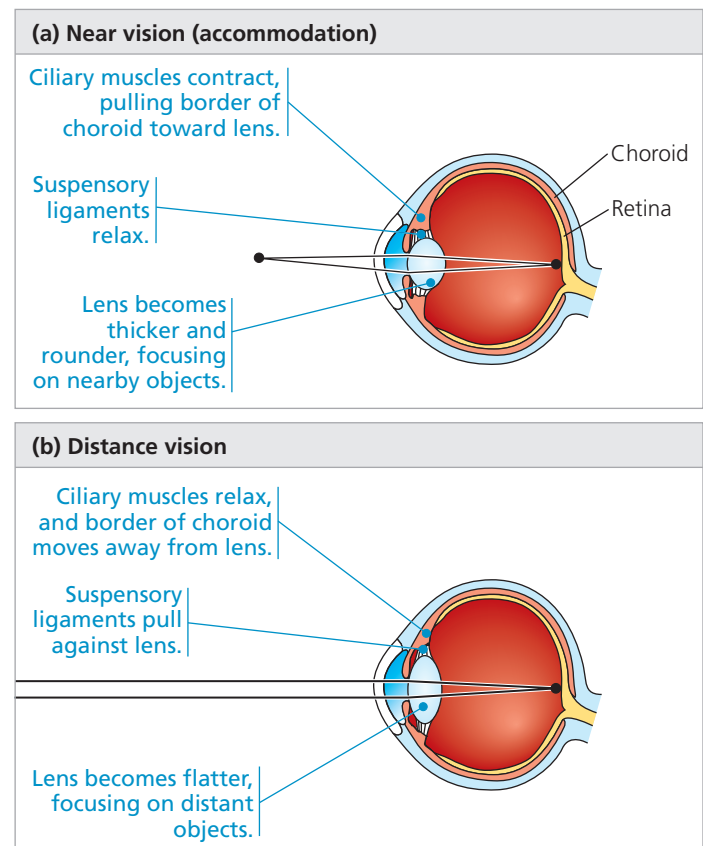


The Visual Field

The brain not only processes visual information but also controls what information is captured. One important control is focusing, which in humans occurs by changing the shape of the lens, as noted earlier and illustrated in **Figure 50.22**. When you focus your eyes on a close object, your lenses become almost spherical. When you view a distant object, your lenses are flattened.

Although our peripheral vision allows us to see objects over a nearly 180° range, the distribution of photoreceptors across the eye limits both what we see and how well we see it. Overall, the human retina contains about 125 million rods and 6 million cones. At the **fovea**, the center of the visual field, there are no rods but a very high density of

▼ **Figure 50.22 Focusing in the mammalian eye.** Ciliary smooth muscles control the shape of the lens, which bends light and focuses it on the retina. The thicker the lens, the more sharply the light is bent.



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